

Activation Functions: Sigmoid, tanh, arctan, ReLU and Softmax

Haiyi Li

Why we need them

A neuron computes a pre-activation and then applies an activation function ϕ :

$$z = w^\top x + b, \quad a = \phi(z). \quad (1)$$

Without ϕ , stacking layers gives nothing new:

$$W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2), \quad (2)$$

which is still one linear map. So depth only helps when ϕ is nonlinear.

For training, ϕ' matters as much as ϕ . Take a chain with one unit per layer, $a_l = \phi(z_l)$ and $z_{l+1} = w_{l+1}a_l$. By the chain rule

$$\frac{\partial a_L}{\partial z_1} = \phi'(z_L) \prod_{l=1}^{L-1} w_{l+1} \phi'(z_l). \quad (3)$$

If $|\phi'|$ is small, this product shrinks with L (vanishing gradient). Below, each function is described by its formula, range, derivative, and how fast ϕ' decays for large $|z|$.

Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad \sigma(z) \in (0, 1), \quad \sigma(0) = \frac{1}{2}. \quad (4)$$

It is symmetric about the point $(0, \frac{1}{2})$:

$$\sigma(-z) = 1 - \sigma(z). \quad (5)$$

The derivative can be written with σ itself:

$$\sigma'(z) = \frac{e^{-z}}{(1 + e^{-z})^2} = \sigma(z)(1 - \sigma(z)), \quad \max_z \sigma'(z) = \sigma'(0) = \frac{1}{4}. \quad (6)$$

For large $|z|$,

$$1 - \sigma(z) = \frac{1}{1 + e^z} \approx e^{-z} \quad (z \rightarrow +\infty), \quad \sigma'(z) \approx e^{-|z|}. \quad (7)$$

Two problems follow. First, with $\sigma' \leq \frac{1}{4}$, (3) gives

$$\left| \frac{\partial a_L}{\partial z_1} \right| \leq \left(\frac{1}{4}\right)^L \prod_{l=1}^{L-1} |w_{l+1}|, \quad \left(\frac{1}{4}\right)^{10} \approx 9.5 \times 10^{-7}. \quad (8)$$

Second, $\sigma(z) > 0$ always. For a unit with inputs $a_j = \sigma(z_j)$ and error signal δ ,

$$\frac{\partial \ell}{\partial w_j} = \delta a_j, \quad (9)$$

so all $\partial\ell/\partial w_j$ share the sign of δ and the weights move in a zig-zag.

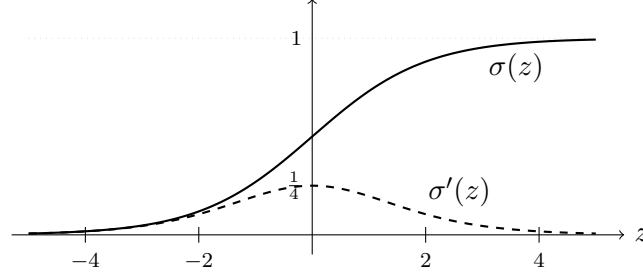


Figure 1: Sigmoid (solid) and its derivative (dashed). The derivative never exceeds $1/4$.

Hyperbolic tangent

$$\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}, \quad \tanh z \in (-1, 1), \quad \tanh 0 = 0. \quad (10)$$

It is a rescaled sigmoid:

$$\tanh z = 2\sigma(2z) - 1, \quad \text{since} \quad 2\sigma(2z) - 1 = \frac{1 - e^{-2z}}{1 + e^{-2z}}. \quad (11)$$

It is odd, so its outputs are centred at 0. The derivative is

$$\tanh' z = 1 - \tanh^2 z = \frac{4}{(e^z + e^{-z})^2}, \quad \max_z \tanh' z = \tanh' 0 = 1, \quad (12)$$

four times the peak of σ' . For large $|z|$,

$$1 - \tanh z = \frac{2}{e^{2z} + 1} \approx 2e^{-2z} \quad (z \rightarrow +\infty), \quad \tanh' z \approx 4e^{-2|z|}, \quad (13)$$

so $\tanh$ saturates even faster than σ . It fixes the sign problem and the small peak, but not saturation.

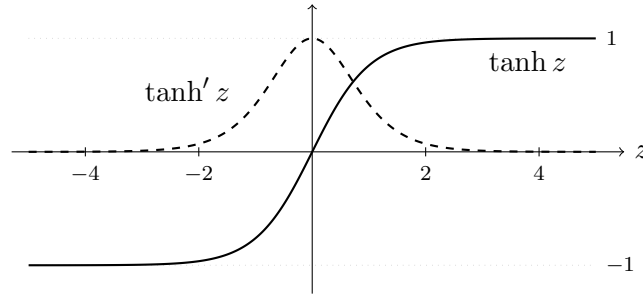


Figure 2: $\tanh$ (solid) and its derivative (dashed). Zero-centred, peak slope 1.

Arctangent

$$\phi(z) = \arctan z, \quad \arctan z \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \arctan 0 = 0. \quad (14)$$

Like $\tanh$ it is odd and zero-centred, with peak slope 1:

$$\frac{d}{dz} \arctan z = \frac{1}{1 + z^2}, \quad \max_z = 1 \text{ at } z = 0. \quad (15)$$

The difference is in the tails. For $z > 0$,

$$\frac{\pi}{2} - \arctan z = \arctan \frac{1}{z} \approx \frac{1}{z}, \quad \frac{1}{1+z^2} \approx \frac{1}{z^2}, \quad (16)$$

so the derivative decays like a power of z , not exponentially. At $z = 5$:

$$\tanh'(5) \approx 1.8 \times 10^{-4}, \quad \sigma'(5) \approx 6.6 \times 10^{-3}, \quad \arctan'(5) = \frac{1}{26} \approx 3.8 \times 10^{-2}. \quad (17)$$

Near 0, $\arctan$ and $\tanh$ are almost the same function:

$$\tanh z = z - \frac{z^3}{3} + \frac{2z^5}{15} - \dots, \quad \arctan z = z - \frac{z^3}{3} + \frac{z^5}{5} - \dots. \quad (18)$$

They agree up to z^3 . All the difference is for large $|z|$. To get the same range as $\tanh$ one can use $\frac{2}{\pi} \arctan z \in (-1, 1)$, whose slope at 0 is $\frac{2}{\pi}$.

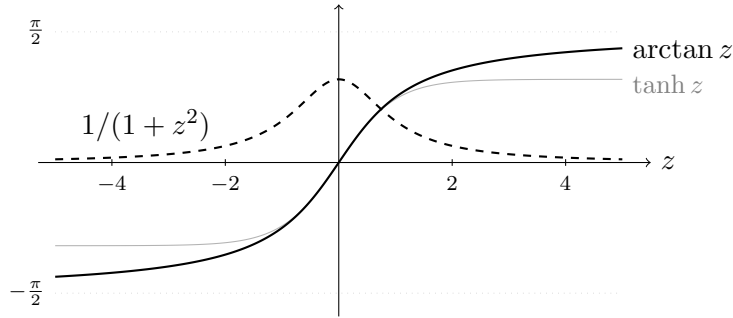


Figure 3: $\arctan$ (solid), its derivative (dashed) and $\tanh$ (grey). The derivative has a long tail.

ReLU

$$\text{ReLU}(z) = \max(0, z), \quad \text{ReLU}(z) \in [0, \infty). \quad (19)$$

The derivative is a step:

$$\text{ReLU}'(z) = \begin{cases} 1, & z > 0, \\ 0, & z < 0, \end{cases} \quad (20)$$

and at $z = 0$ any value in $[0, 1]$ is a subgradient; libraries use 0.

On active units ($z_l > 0$) every factor $\phi'(z_l)$ in (3) equals 1:

$$\frac{\partial a_L}{\partial z_1} = \prod_{l=1}^{L-1} w_{l+1} \quad \text{if all } z_l > 0, \quad (21)$$

so the activation itself causes no vanishing. Other properties:

$$\text{ReLU}(cz) = c \text{ReLU}(z) \quad (c \geq 0), \quad \text{ReLU}(z) = 0 \text{ for } z < 0. \quad (22)$$

The first is positive homogeneity. The second gives sparse outputs and costs one comparison, but a unit with $z < 0$ for every input gets zero gradient forever (a “dead” unit). Leaky ReLU keeps a small slope α (often 0.01) on the negative side:

$$\text{LReLU}_\alpha(z) = \max(\alpha z, z), \quad \text{LReLU}'_\alpha(z) = \begin{cases} 1, & z > 0, \\ \alpha, & z < 0. \end{cases} \quad (23)$$

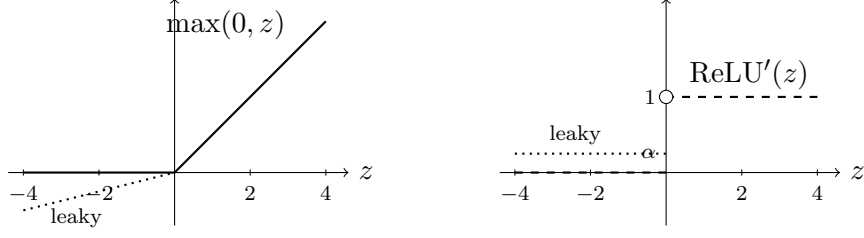


Figure 4: Left: ReLU (solid) and leaky ReLU (dotted). Right: their derivatives. $\alpha = 0.25$ here for visibility.

Softmax

Softmax acts on a whole vector $z \in \mathbb{R}^K$, not on one number:

$$s_i(z) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad i = 1, \dots, K. \quad (24)$$

The output is a probability vector:

$$s_i > 0, \quad \sum_{i=1}^K s_i = 1. \quad (25)$$

Adding the same constant to every entry changes nothing:

$$s(z + c\mathbf{1}) = s(z), \quad (26)$$

so in practice one computes $s(z - \max_j z_j)$ to avoid overflow in e^{z_i} .

The Jacobian couples all entries:

$$\frac{\partial s_i}{\partial z_j} = s_i(\delta_{ij} - s_j), \quad J = \text{diag}(s) - s s^\top. \quad (27)$$

With $K = 2$ softmax is the sigmoid of the difference:

$$s_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-(z_1 - z_2)}} = \sigma(z_1 - z_2). \quad (28)$$

With the cross-entropy loss $\ell = -\sum_i y_i \log s_i$ and a one-hot label y , the gradient is simply

$$\frac{\partial \ell}{\partial z_j} = -\sum_i y_i(\delta_{ij} - s_j) = s_j - y_j, \quad \nabla_z \ell = s - y. \quad (29)$$

A temperature $T > 0$ controls how sharp the output is:

$$s(z/T) \longrightarrow \begin{cases} \text{one-hot at } \arg \max_i z_i, & T \rightarrow 0, \\ (\frac{1}{K}, \dots, \frac{1}{K}), & T \rightarrow \infty. \end{cases} \quad (30)$$

Because the units are coupled, softmax is used at the output layer for classification, not as a hidden-layer activation.

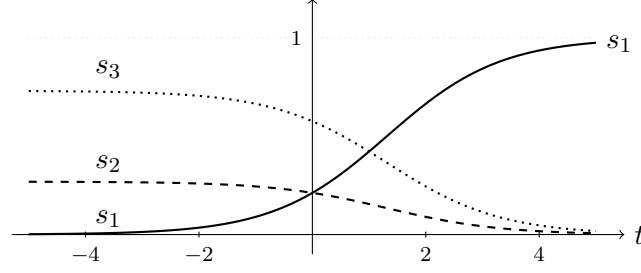


Figure 5: Softmax of $z = (t, 0, 1)$ as t varies: s_1 (solid), s_2 (dashed), s_3 (dotted). The three curves always add up to 1.

Comparison

The four scalar functions on one plot, then their derivatives.

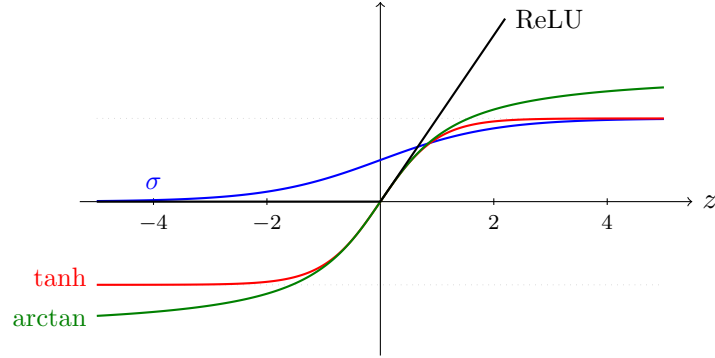


Figure 6: σ (blue), $\tanh$ (red), $\arctan$ (green), ReLU (black).

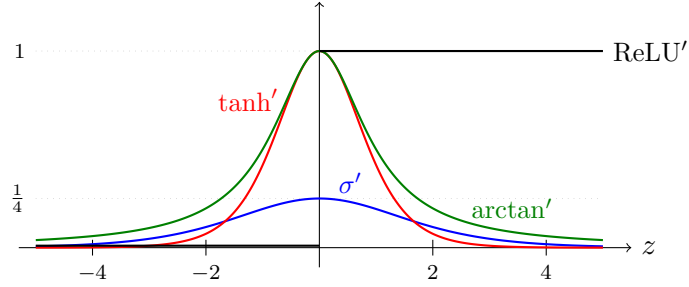


Figure 7: Derivatives. Peaks: $\frac{1}{4}$ for σ , 1 for the others. Only $\arctan'$ keeps a visible tail at $|z| = 5$.

	range	$\phi(0)$	$\phi'(0)$	zero-centred	$\phi'(z)$ for large $ z $
σ	$(0, 1)$	$\frac{1}{2}$	$\frac{1}{4}$	no	$\approx e^{- z }$
$\tanh$	$(-1, 1)$	0	1	yes	$\approx 4e^{-2 z }$
$\arctan$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	0	1	yes	$\approx z^{-2}$
ReLU	$[0, \infty)$	0	—	no	1 ($z > 0$), 0 ($z < 0$)
softmax	simplex	$\frac{1}{K}\mathbf{1}$	$\text{diag}(s) - ss^\top$	—	—

Table 1: Summary.

How the functions are related:

$$\tanh z = 2\sigma(2z) - 1, \quad \text{see (11),} \quad (31)$$

$$\sigma(z_1 - z_2) = \text{softmax}(z_1, z_2)_1, \quad \text{see (28),} \quad (32)$$

$$\tanh z - \arctan z = -\frac{z^5}{15} + O(z^7), \quad \text{from (18).} \quad (33)$$

Ordering by how fast the gradient dies for large $|z|$:

$$\underbrace{4e^{-2|z|}}_{\tanh'} \ll \underbrace{e^{-|z|}}_{\sigma'} \ll \underbrace{z^{-2}}_{\arctan'}, \quad \text{ReLU}' = 1 \text{ for } z > 0. \quad (34)$$

In short:

- hidden layers: ReLU (or leaky ReLU) by default; no saturation for $z > 0$, cheap;
- tanh: when a bounded, zero-centred output is needed, e.g. recurrent states;
- σ : gates and binary outputs, where a value in $(0, 1)$ is wanted;
- arctan: same shape as tanh near 0 with a slower-decaying gradient, rarely used in practice;
- softmax: output layer for K -class classification, paired with cross-entropy so that $\nabla_z \ell = s - y$.